

Binary LP

$$\begin{aligned} \min \quad & c^T x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0 \\ & x \in \{0, 1\}^q \end{aligned}$$

EX Vertex Cover

$$\begin{aligned} \dots \\ \text{s.t.} \quad & x_i + x_j \geq 1 \\ \dots \\ & x \in \{0, 1\}^q \end{aligned}$$

Having an “And” variable

$$\begin{aligned} x \wedge y \rightarrow z (= xy) (= x \wedge y) \\ x, y \geq z \geq x + y - 1 \end{aligned}$$

This allows for $z = 1$ iff x and y are 1 and 0 otherwise. This creates the flexibility for this constraint to be true without forcing $x, y = 1$

Misc

$$\begin{aligned} Q \in R^{q \times q}, x \in R^p \\ x^T Q x = \sum_i^p \sum_j^q q_{ij} x_i x_j \end{aligned}$$

This is an arbitrary homogenous quadratic form.

EX

$$Q = [(1, 2), (3, 4)] \rightarrow x^T Q x = x_1^2 + 2x_1x_2 + 3x_1x_2 + 4x_2^2$$

We can assume Q is symmetric because it does not change the $x^T Q x$